DFT calculation

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1 Destabilisation of MgH_2 by substitutional Ni and Co dopants: a first-principles study

Pure MgH_2 stores 7.6 % hydrogen by mass, but its formation enthalpy ($\Delta H_f \approx -75\,\mathrm{kJ\,mol^{-1}}$ H_2, Bogdanović et al. [1]) is so negative – the hydride so stable – that release requires temperatures around 300 °C. A useful catalyst for this system is one that destabilises the hydride – makes ΔH less negative – so that release happens at lower temperature. Nickel in particular is known to improve the hydrogen sorption properties of MgH_2 [2]. To provide a first-principles rationale for the Ni- and Co-doped magnesium hydrides investigated experimentally in this work, we carried out density-functional theory (DFT) calculations of the hydride formation enthalpy. Within a single internally consistent setup we quantify whether substitutional Ni and Co at the Mg site destabilise the hydride at a fixed dopant concentration.

1.1 Methods

We compare three hydrogenation pathways on a common per-H_2 basis:

$$\mathrm{Mg} + \mathrm{H_2} \longrightarrow \mathrm{MgH_2} \qquad \text{(pure)}$$
$$\mathrm{Mg_{14}Ni_2} + 16\,\mathrm{H_2} \longrightarrow \mathrm{Mg_{14}Ni_2H_{32}} \qquad \text{(Ni dopant, 12.5\%)}$$
$$\mathrm{Mg_{14}Co_2} + 16\,\mathrm{H_2} \longrightarrow \mathrm{Mg_{14}Co_2H_{32}} \qquad \text{(Co dopant, 12.5\%)}$$

The two doped pathways use the same supercell topology – a 2×2×2 replication of the rutile MgH_2 cell for the products and the matching $2\times2\times2$ hexagonal close-packed Mg cell for the reactants, with two of the sixteen metal sites occupied by the dopant X=Ni,Co, i.e. 12.5 atomic percent (at.%) of the metal atoms. The two doped pathways differ only in the chemical identity of the dopant, and each is referenced to the same pristine pathway, so the destabilisation $\Delta\Delta H$ relative to pure MgH_2 benefits from cancellation of supercell-size, k-point-sampling, and symmetry errors. Figure 1 shows the three relaxed product structures.

All calculations are performed with Quantum ESPRESSO [3]. The exchange-correlation functional is the Perdew–Burke–Ernzerhof (PBE) generalised-gradient approximation, augmented with Grimme-D3 dispersion [4]. Plane-wave cutoffs are 60 Ry (wavefunctions) and 480 Ry (charge density). Pseudopotentials are of the projector augmented-wave (PAW) type for Mg and H and Vanderbilt ultrasoft pseudopotentials (USPP) for Ni and Co. Brillouin-zone sampling uses Γ-centred Monkhorst–Pack grids of $12\times12\times8$ for the Mg primitive, $8\times8\times12$ for rutile MgH_2, $6\times6\times4$ for the Mg supercells, $4\times4\times6$ for the hydride supercells, and the Γ point for the isolated H_2 molecule. Metallic phases are described with cold smearing of width 0.01 Ry, while the isolated H_2 molecule uses fixed occupations. The self-consistent field (SCF) cycle is converged to 1×10^{-9} Ry using Anderson charge-density mixing. Cell parameters and atomic positions are relaxed simultaneously with a quasi-Newton scheme until the residual forces fall below $1 \times 10^{-4}\,\mathrm{Ry}\,{a_0}^{-1}$. For the doped supercells no crystal symmetry is imposed during relaxation,

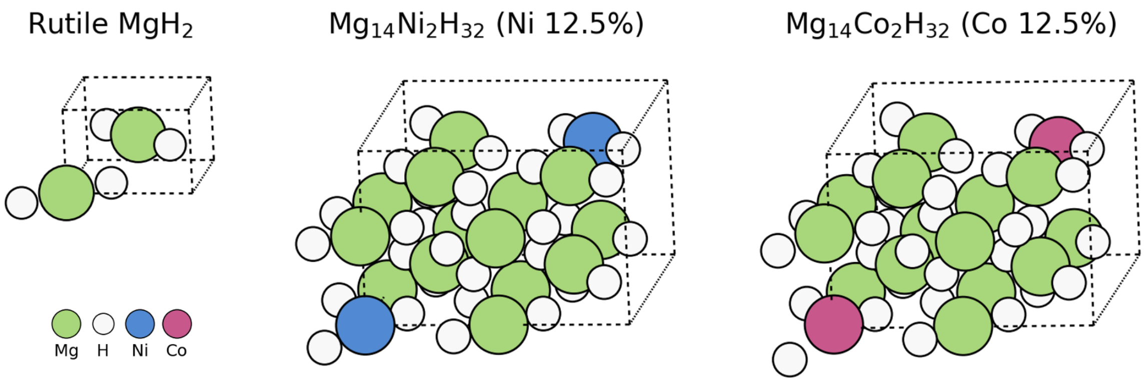

Figure 1: Relaxed product structures: pristine rutile MgH_2 (left), $Mg_{14}Ni_2H_{32}$ (centre), and $Mg_{14}Co_2H_{32}$ (right), the latter two at 12.5 at.% substitution. In the doped cells the two transition-metal atoms occupy diagonally opposite Mg sites in the $2\times2\times2$ rutile supercell, maximising their separation to minimise dopant–dopant interaction.

so that the small unit-cell drift accumulated over the optimisation does not interfere with the symmetry analysis.

For each pathway the per-H_2 formation enthalpy is

$$\Delta H = \frac{1}{n_{\mathrm{H}_2}} \left[E(\text{hydride cell}) \; - \; E(\text{metal cell}) \; - \; n_{\mathrm{H}_2}\, E(\mathrm{H}_2) \right] \times 1312.75\,\mathrm{kJ\,mol^{-1}\,Ry^{-1}}. \quad (1)$$

We use a non-spin-polarised treatment throughout. A spin-polarised check on $Mg_{14}Ni_2$ gave $0\,\mu_\mathrm{B}$ per Ni, so spin polarisation has no effect on the Ni pathway, and in test runs its influence on ΔH was below $1\,\mathrm{kJ\,mol^{-1}}$ H_2. Cobalt carries a moment of $\sim 1.3\,\mu_\mathrm{B}$ per atom in the metallic reactant that is largely quenched by H bonding in the hydride; treating reactant and product on the same non-magnetic footing is expected to cancel most of the resulting bias in ΔH, and it also removes a spin-related convergence instability that otherwise prevented the SCF cycle from converging for $Mg_{14}X_2H_{32}$. The residual spin correction for the Co pathway has not been quantified and is the main uncertainty specific to that pathway, although given its magnitude relative to $\Delta\Delta H$ it is not expected to alter the qualitative ordering.

1.2 DFT results

All seven calculations completed successfully, with the SCF cycle converged at every step of the relaxation. The geometry optimisation reached $|\boldsymbol{F}| < 5 \times 10^{-4}\,\mathrm{Ry\,a_0}^{-1}$ on six structures; for $Mg_{14}Ni_2H_{32}$ the relaxation stalled at $|\boldsymbol{F}| = 5 \times 10^{-3}\,\mathrm{Ry\,a_0}^{-1}$, but its total energy was already converged to within one part in 10^6, so the residual numerical uncertainty is $\lesssim 1\,\mathrm{kJ\,mol^{-1}}$ H_2. The resulting formation enthalpies are summarised in Table 1.

Pathway	ΔH / $\mathrm{kJ\,mol^{-1}}$ H_2	$\Delta\Delta H$ vs. pure
pure Mg	–63.75	—
Ni 12.5% $Mg_{14}Ni_2$	–48.80	14.95
Co 12.5% $Mg_{14}Co_2$	–55.24	8.51

Table 1: Formation enthalpies of magnesium hydride per absorbed H_2 for the three pathways. Less-negative ΔH means the hydride is less stable, i.e. easier to dehydrogenate – the desired catalyst effect.

The pure-Mg baseline ($\Delta H = -63.75\,\mathrm{kJ\,mol^{-1}}\ \mathrm{H_2}$) sits about $11\,\mathrm{kJ\,mol^{-1}}$ above experiment ($-75\,\mathrm{kJ\,mol^{-1}}\ \mathrm{H_2}$ [1]), reflecting the known PBE under-binding of hydrides [5]. The inter-pathway differences $\Delta\Delta H$ are differential quantities and benefit from cancellation of this systematic under-binding error, so they are meaningful even at the absolute level of accuracy delivered here.

Table 1 shows that both Ni and Co dopants at 12.5 at.% destabilise MgH_2 relative to the pristine pathway: both $\Delta\Delta H$ values are positive and exceed by an order of magnitude the residual numerical uncertainty of the calculation. Each dopant therefore makes the hydride easier to dehydrogenate than pure MgH_2. This is consistent with the long-standing experimental finding that Ni-containing magnesium hydrides are less stable than pure MgH_2: the dedicated ternary phase Mg_2NiH_4 desorbs with $\Delta H \approx -64.4\,\mathrm{kJ\,mol^{-1}}\ \mathrm{H_2}$ [6], some $10\,\mathrm{kJ\,mol^{-1}}$ less stable than MgH_2. We emphasise that this is a comparison of trend only: our $Mg_{14}Ni_2H_{32}$ models dilute (12.5 at.%) substitution within the rutile MgH_2 lattice, which is a distinct system from the ordered, Ni-rich Mg_2NiH_4 phase, so the absolute enthalpies are not expected to coincide.

In this thermodynamic sense a catalyst weakens the H–metal bond just enough to lower the release temperature without sacrificing storage capacity, and substitutional Ni or Co does exactly that. These calculations thus provide a thermodynamic rationale for the improved dehydrogenation observed for the Ni- and Co-doped samples in this work.

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